DAY 2

# Probability

## *& Statistics Basics*

**Theory:** 1 hour **Practical:** 2 hours

Topics: Data types · Descriptive stats · Probability · Distributions

LESSON OVERVIEW

# What We'll Learn Today

1

0:00 – 1:00

## Theory

Data types, stats, probability, distributions

2

1:10 – 2:10

## Practical 1

Worked examples — compute by hand

3

2:20 – 3:00

## Practical 2

Pair practice + real-world problems

💡 *Probability shows up EVERYWHERE in data science and ML — let's go!*

PAR

# Theory • 1 hour

*From raw data to making predictions*

# Why do we need statistics?

***Statistics helps us MAKE SENSE of data. Probability helps us PREDICT what comes next.***

**Everyday examples:**

### Weather

70% chance of rain tomorrow

### Medicine

This drug works in 85% of patients

### Spam filter

Is this email spam or not?

### ML models

How confident is the prediction?

# Before we analyze, we classify

*Data comes in two big families:*

## CATEGORICAL

*Data in groups / labels (not numbers)*

### Nominal

No natural order. E.g., colors (red / blue), gender, country names.

### Ordinal

HAS order, but gaps aren't equal. E.g., small/medium/large, ratings 1–5.

## NUMERICAL

*Data that is actual numbers (you can do math on it)*

### Discrete

Whole-number counts. E.g., #students in class, #goals in a match.

### Continuous

Any value in a range. E.g., height (1.73 m), weight, temperature.

QUICK CHECK

# Can you classify these?

| | |
|---|---|
| Blood type (A, B, AB, O)<br>Nominal | Ranking in a race (1st, 2nd, 3rd)<br>Ordinal |
| Number of siblings<br>Discrete | Height in centimetres<br>Continuous |
| T-shirt size (S, M, L, XL)<br>Ordinal | Favourite movie genre<br>Nominal |

# Summarising Data in a Few Numbers

*When you have 1,000 numbers, you don't read them all — you SUMMARISE them.*

## CENTRAL TENDENCY

*Where's the middle?*

### Mean

The average — add them all, divide by count.

### Median

The middle value when you line them up in order.

### Mode

The most common value (appears most often).

## SPREAD / VARIABILITY

*How spread out is the data?*

### Variance

The average squared distance from the mean.

### Standard Deviation

Square root of variance — same units as the data.

### Range

Simplest: max − min (we'll only mention it).

# Add them all, divide by how many

$$\text{mean} = (x_1 + x_2 + \ldots + x_n) / n$$

Worked Example

**Test scores: 7, 8, 10, 6, 9**

Step 1: Add them up: $7 + 8 + 10 + 6 + 9 = 40$

Step 2: Count how many numbers there are: $n = 5$

Step 3: Divide the sum by the count: $40 / 5 = 8$

**Answer: Mean = 8**

# The Other Two "Middles"

## MEDIAN

*The middle value after sorting*

**Example (odd count):**

Data: 3, 7, 5, 9, 4

Sort: 3, 4, 5, 7, 9

Middle = 5

---

*Even count? Average the two middle numbers:*

Data: 2, 3, 6, 8 → $(3 + 6)/2 = 4.5$

🔑 **Median is NOT affected by extreme values (outliers).**

## MODE

*The value that appears most often*

**Example:**

Shoe sizes: 8, 9, 7, 9, 10, 9, 8

*Count each:*

7 → 1 time, 8 → 2 times, 9 → 3 times, 10 → 1 time

Mode = 9

🔑 **Mode is the ONLY middle you can use for categorical data.**

# How Spread Out is the Data?

**Variance $\sigma^2 = \Sigma (x_i - \mu)^2 / n$    Std. Dev. $\sigma = \sqrt{\text{Variance}}$**

*In words: how FAR each value is from the mean, on average (squared, then averaged).*

### Worked Example

**Data: 2, 4, 4, 6, Mean $\mu = 4$**

Step 1 — deviations $(x_i - \mu)$: $2-4 = -2$, $4-4 = 0$, $4-4 = 0$, $6-4 = 2$

Step 2 — square each: $(-2)^2 = 4$, $0^2 = 0$, $0^2 = 0$, $2^2 = 4$

Step 3 — sum the squares: $4 + 0 + 0 + 4 = 8$

Step 4 — divide by n = 4: $\sigma^2 = 8 / 4 = 2$ (Variance)

Step 5 — square root: $\sigma = \sqrt{2} \approx 1.41$ (Std. Deviation)

# The Rules of Chance

P(event) = (favourable outcomes) / (total outcomes)

## Sample Space

All possible outcomes. Written as S.

E.g., rolling a die: S = {1, 2, 3, 4, 5, 6}

## Event

A SUBSET of the sample space (what we want).

E.g., "rolling even": E = {2, 4, 6}

## Probability

A number between 0 and 1.

0 = impossible, 1 = certain,
0.5 = equal chance.

QUICK EXAMPLES

# Coins and Dice

## Flipping a FAIR coin

Sample space: S = { Heads, Tails }

Event "get Heads": E = { Heads }

**P(Heads) = 1 / 2 = 0.5 = 50%**

*Interpretation: if you flip 1,000 coins, about 500 should be Heads.*

## Rolling a FAIR 6-sided die

Sample space: S = {1, 2, 3, 4, 5, 6}

Event "roll even": E = {2, 4, 6}

**P(even) = 3 / 6 = 0.5 = 50%**

*Event "roll a 7": impossible (not in S) → P = 0.*

# When we want A OR B

### MUTUALLY EXCLUSIVE

*Events that CAN'T happen together*

$$P(A \cup B) = P(A) + P(B)$$

**Example:**

Rolling a die — P(1 or 6)?

*The die can't show both at once.*

$= P(1) + P(6) = 1/6 + 1/6 = 2/6 = 1/3$

🔑 **If events can't overlap → just add probabilities.**

### NOT MUTUALLY EXCLUSIVE

*Events that CAN overlap*

$$P(A \cup B) = P(A) + P(B) - P(A \cap B)$$

**Example:**

Card — P(Heart or Face card)?

*A card can be BOTH a heart AND a face card.*

$= 13/52 + 12/52 - 3/52 = 22/52 \approx 0.42$

🔑 **Subtract the overlap, so you don't count it twice.**

# When we want A AND B

## INDEPENDENT EVENTS

*One doesn't affect the other*

$$P(A \cap B) = P(A) \cdot P(B)$$

**Example:**

Flip 2 coins — P(both Heads)?

*The 1st coin doesn't affect the 2nd.*

$= P(H) \cdot P(H) = 1/2 \cdot 1/2 = 1/4$

🔑 **Independent = outcome of A doesn't change P(B).**

## DEPENDENT EVENTS

*One DOES affect the other*

$$P(A \cap B) = P(A) \cdot P(B \mid A)$$

**Example:**

Draw 2 cards without replacement:

*P(both Aces)? Deck shrinks after 1st draw.*

$= 4/52 \cdot 3/51 \approx 0.0045$

🔑 **Dependent = use CONDITIONAL probability P(B|A).**

# "Given that..." — updating our beliefs

$$P(A \mid B) = P(A \cap B) \; / \; P(B)$$

*Read "|" as "given that". P(A | B) = "probability of A, given B already happened."*

### Worked Example

**Roll a die. Given the result is EVEN, what's the probability it's a 4?**

Step 1: Let A = "roll a 4", B = "roll even".

Step 2: $B = \{2, 4, 6\}$, so $P(B) = 3/6$.

Step 3: $A \cap B = \{4\}$ (the overlap), so $P(A \cap B) = 1/6$.

Step 4: $P(A \mid B) = P(A \cap B) \; / \; P(B) = (1/6) \; / \; (3/6) = 1/3$

**Answer: $P(4 \mid \text{even}) = 1/3 \approx 33\%$**

# The Famous "Bell Curve"

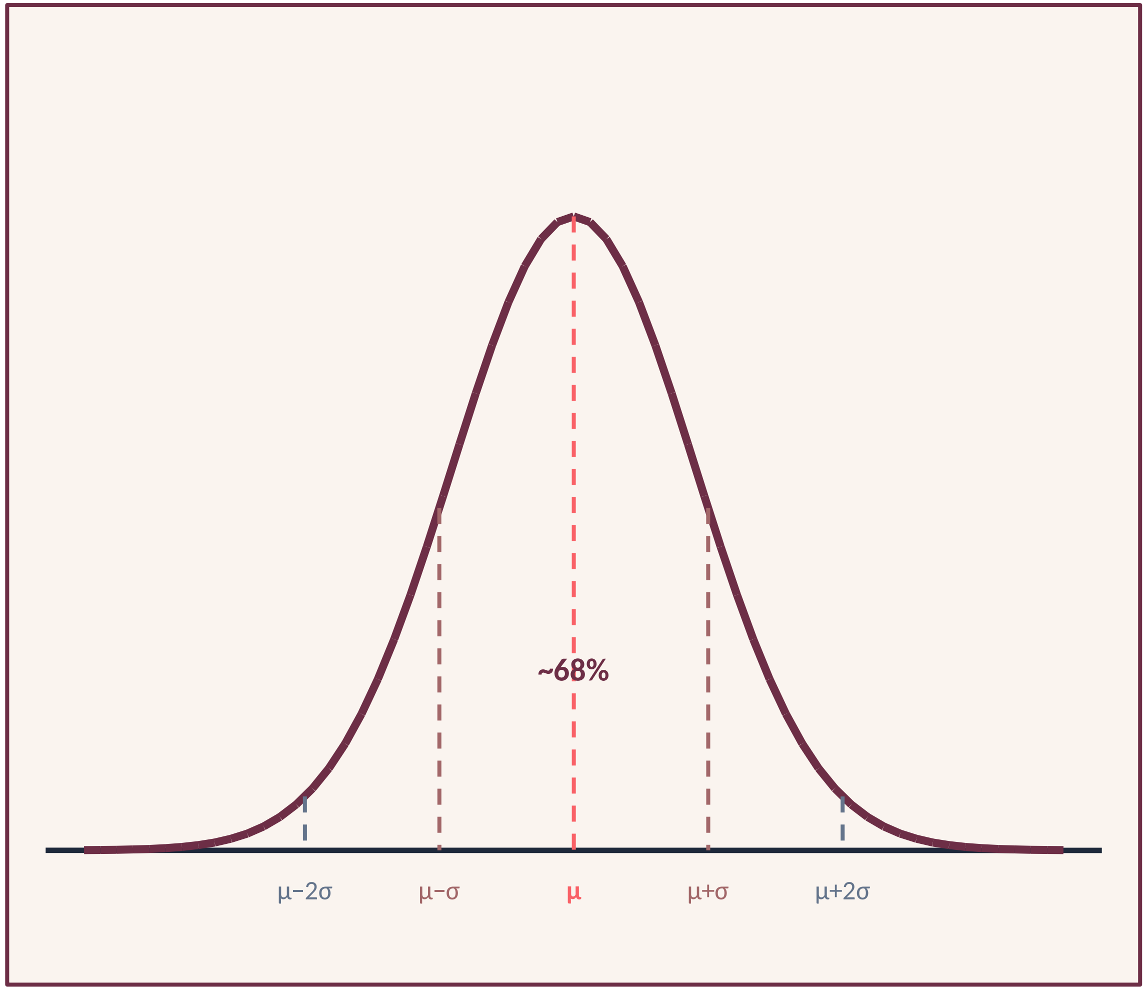
**The most important distribution in statistics.**

*Shows up everywhere — heights, test scores, measurement errors, noise in data...*

**Key properties:**

- Symmetric — same on both sides of the mean.
- Mean = median = mode (all at centre).
- Defined by two numbers: μ (mean) and σ (std. dev.).
- Most values are near the mean; extremes are rare.

🔑 **68 – 95 – 99.7 rule: ~68% within 1σ, ~95% within 2σ, ~99.7% within 3σ.**



# Binomial & Uniform (high-level)

## BINOMIAL

*Counting successes in yes/no trials*

**When to use:**

- Fixed number of trials (n)
- Each trial is independent
- Only 2 outcomes (success/failure)
- Same probability p every time

*Example: Flip a coin 10 times → count Heads.*

*Example: 100 emails → count how many are spam.*

*Shape: bars — a histogram (discrete).*

## UNIFORM

*Every outcome is equally likely*

**When to use:**

- Fair randomness (no preferred outcome)
- All values in a range have same chance
- Often the DEFAULT when you know nothing

*Example: Fair die — every face has P = 1/6.*

*Example: Random number between 0 and 1.*

*Shape: a flat rectangle.*

# Theory — What We Learned

1

**Data has 4 types**

Nominal, ordinal, discrete, continuous

2

**Summarize with stats**

Mean, median, mode + variance & std. dev.

3

**Probability = favourable / total**

Addition for OR, multiplication for AND

4

**Distributions describe shapes**

Normal (bell), binomial (counts), uniform (flat)

☕ ***Break time — see you in 10 minutes, then we PRACTICE!***

P A R

# Practical • 2 hours

*Let's compute with real numbers*

PRACTICAL · EXAMPLE 1

# Worked Example (Together on the Board)

**Question:**

**Find the mean, median, and mode of: 4, 8, 6, 4, 10**

**Solution:**

| | |
|---|---|
| **Sort:** | Order first: 4, 4, 6, 8, 10 |
| **Mean:** | (4 + 4 + 6 + 8 + 10) / 5 = 32 / 5 = 6.4 |
| **Median:** | Middle of 5 numbers = the 3rd = 6 |
| **Mode:** | Most frequent value = 4 (appears twice) |

**Answer: Mean = 6.4, Median = 6, Mode = 4**

PRACTICAL · EXAMPLE 2

# Worked Example (Together on the Board)

**Question:**

**Find the variance and standard deviation of: 3, 5, 7, 9**

**Solution:**

| | |
|---|---|
| **Step 1:** | Mean $\mu = (3 + 5 + 7 + 9) / 4 = 24/4 = 6$ |
| **Step 2:** | Deviations: $3-6=-3$, $5-6=-1$, $7-6=1$, $9-6=3$ |
| **Step 3:** | Square them: 9, 1, 1, 9 |
| **Step 4:** | Sum: $9+1+1+9 = 20$. Divide by $n=4$: $\sigma^2 = 20/4 = 5$ |
| **Step 5:** | $\sigma = \sqrt{5} \approx 2.24$ |

**Answer: Variance = 5, Std. Deviation ≈ 2.24**

# Worked Example (Together on the Board)

**Question:**

**A bag has 3 red, 5 blue, and 2 green balls. Pick one at random. Find P(blue).**

**Solution:**

**Step 1:** Total balls = 3 + 5 + 2 = 10

**Step 2:** Favourable outcomes = 5 (the blue balls)

**Step 3:** P(blue) = favourable / total = 5 / 10

**Step 4:** Simplify = 1/2 = 0.5

**Answer: P(blue) = 0.5 = 50%**

PRACTICAL · EXAMPLE 4

# Worked Example (Together on the Board)

**Question:**

**Roll a fair die. What's $P(\text{rolling a 2 OR a 5})$?**

**Solution:**

**Step 1:** Can a single roll be both 2 AND 5? No → mutually exclusive.

**Step 2:** Use simple addition rule: $P(A \cup B) = P(A) + P(B)$

**Step 3:** $P(2) = 1/6$, $P(5) = 1/6$

**Step 4:** Add: $1/6 + 1/6 = 2/6$

**Step 5:** Simplify: $2/6 = 1/3 \approx 0.333$

**Answer: $P(2 \text{ or } 5) = 1/3 \approx 33.3\%$**

PRACTICAL · EXAMPLE 5

# Worked Example (Together on the Board)

Question:

**Flip a coin and roll a die. What's $P(\text{Heads AND roll a 6})$?**

Solution:

**Step 1:** Coin and die don't affect each other → INDEPENDENT.

**Step 2:** Use: $P(A \cap B) = P(A) \cdot P(B)$

**Step 3:** $P(\text{Heads}) = 1/2, \quad P(6) = 1/6$

**Step 4:** Multiply: $1/2 \cdot 1/6 = 1/12$

**Step 5:** As decimal: $\approx 0.083$

**Answer: $P(\text{Heads and } 6) = 1/12 \approx 8.3\%$**

# Worked Example (Together on the Board)

**Question:**

**In a class of 30: 18 like math, 12 like coding, 8 like BOTH. Pick a student who likes coding. What's P(they also like math)?**

**Solution:**

| | |
|---|---|
| **Step 1:** | We need P(math \| coding). Let A = math, B = coding. |
| **Step 2:** | P(coding) = 12/30 = 0.4 |
| **Step 3:** | P(math AND coding) = 8/30 ≈ 0.267 |
| **Step 4:** | $P(A \mid B) = P(A \cap B) / P(B) = (8/30) / (12/30)$ |
| **Step 5:** | The /30 cancels: = 8/12 = 2/3 ≈ 0.667 |

**Answer: P(math | coding) = 2/3 ≈ 66.7%**

PRACTICAL 2 · PAIR WORK

# Your Turn — Work in Pairs

## 👥 How this works

- Pair up with the person next to you.
- Work through the practice questions on the next slides.
- Explain your steps to your partner out loud.
- Stuck? Raise your hand — we'll help!
- After ~30 min, we'll review every answer together.

📝 ***Calculator allowed — but please show your working!***

P R A C T I

# 10 Practice Questions

*Solutions follow — try them FIRST, then check!*

PRACTICE QUESTIONS

# Questions 1 – 5

Q1

**Classify this data: "T-shirt sizes (S, M, L, XL)"**

💡 *Hint: Categorical with order?*

Q2

**Find the mean, median, and mode of: 5, 7, 3, 7, 8**

💡 *Hint: Sort first*

Q3

**Find the variance of: 2, 4, 6**

💡 *Hint: Mean first, then squared deviations*

Q4

**Pick a random card from a standard 52-card deck. P(Queen)?**

💡 *Hint: Count the Queens*

Q5

**A spinner has 4 equal sections colored Red, Blue, Green, Yellow. P(Red OR Green)?**

💡 *Hint: Mutually exclusive → add*

PRACTICE QUESTIONS

# Questions 6 – 10

**Q6** **Flip a fair coin 3 times. P(all 3 are Heads)?**

💡 *Hint: Independent → multiply*

**Q7** **Draw 2 cards without replacement. P(both are Hearts)?**

💡 *Hint: Dependent — deck shrinks*

**Q8** **Roll a die. Given the result is greater than 3, what's P(it's a 6)?**

💡 *Hint: Conditional: $P(A|B) = P(A \cap B)/P(B)$*

**Q9** **Exam scores are Normal with $\mu = 70$, $\sigma = 10$. Roughly what % score between 60 and 80?**

💡 *Hint: 68-95-99.7 rule; that's $\pm 1\sigma$*

**Q10** **A spam filter catches 90% of spam. If 200 spam emails arrive, about how many will it catch?**

💡 *Hint: Expected value = $n \cdot p$*

S O L U

# Answer Key

*Check your work — and understand every step!*

SOLUTION

# Q1: Data Classification

**Problem: Classify "T-shirt sizes (S, M, L, XL)".**

**Step-by-step:**

- It's not numbers, so it's CATEGORICAL.
- Inside categorical, ask: is there a natural order?
- S < M < L < XL — yes, there IS order.
- But gaps aren't equal (is L−M the same "jump" as XL−L? Not clearly).
- So it's ORDINAL (categorical with order).

**✓ Answer: Ordinal (categorical, ordered)**

SOLUTION

# Q2: Mean, Median, Mode

**Problem: Find all three for: 5, 7, 3, 7, 8**

**Step-by-step:**

- ▶ Sort: 3, 5, 7, 7, 8
- ▶ MEAN = (3 + 5 + 7 + 7 + 8) / 5 = 30 / 5 = 6
- ▶ MEDIAN = middle of 5 numbers = 3rd number = 7
- ▶ MODE = most frequent value = 7 (appears twice)

**✓ Answer: Mean = 6, Median = 7, Mode = 7**

SOLUTION

# Q3: Variance

**Problem: Find the variance of: 2, 4, 6**

**Step-by-step:**

- Mean $\mu = (2 + 4 + 6) / 3 = 12/3 = 4$
- Deviations from mean: $2-4 = -2$, $4-4 = 0$, $6-4 = 2$
- Square each: 4, 0, 4
- Sum of squares: $4 + 0 + 4 = 8$
- Divide by $n = 3$: $\sigma^2 = 8 / 3 \approx 2.67$

**✓ Answer: Variance $\sigma^2 \approx 2.67$**

SOLUTION

# Q4: Simple Probability

**Problem: P(Queen) from a standard 52-card deck**

**Step-by-step:**

- How many Queens are in a deck? 4 (one per suit).
- Total cards = 52.
- P(Queen) = favourable / total = 4 / 52
- Simplify: 4/52 = 1/13 ≈ 0.077

**✓ Answer: P(Queen) = 1/13 ≈ 7.7%**

SOLUTION

# Q5: Addition Rule — Mutually Exclusive

**Problem: Spinner: R, B, G, Y (equal). P(Red OR Green)?**

**Step-by-step:**

- Can a single spin land on both Red AND Green? No → mutually exclusive.
- Use: $P(A \cup B) = P(A) + P(B)$
- $P(\text{Red}) = 1/4$, $P(\text{Green}) = 1/4$
- Add: $1/4 + 1/4 = 2/4 = 1/2$

**✓ Answer: P(Red or Green) = 1/2 = 50%**

SOLUTION

# Q6: Multiplication Rule — Independent

**Problem: Flip a fair coin 3 times. P(all 3 Heads)?**

**Step-by-step:**

- Each flip is INDEPENDENT (previous flip doesn't affect next).
- Use: $P(A \cap B \cap C) = P(A) \cdot P(B) \cdot P(C)$
- $P(H) = 1/2$ for each flip.
- Multiply: $1/2 \cdot 1/2 \cdot 1/2 = 1/8$
- As decimal: $= 0.125$

**✓ Answer: $P(HHH) = 1/8 = 12.5\%$**

SOLUTION

# Q7: Dependent Events

**Problem: Draw 2 cards without replacement. P(both Hearts)?**

**Step-by-step:**

- Draw is dependent — deck changes after first card.
- P(1st card is Heart) = 13/52 = 1/4
- After drawing 1 Heart: 12 Hearts left, 51 cards total.
- P(2nd Heart | 1st was Heart) = 12/51
- Multiply: (13/52) · (12/51) = 156/2652 ≈ 0.0588

**✓ Answer: P(both Hearts) ≈ 0.0588 ≈ 5.9%**

SOLUTION

# Q8: Conditional Probability

**Problem: Roll a die. Given result > 3, P(it's a 6)?**

**Step-by-step:**

- ▸ Let A = "roll a 6", B = "result > 3".
- ▸ $B = \{4, 5, 6\} \to P(B) = 3/6$
- ▸ $A \cap B = \{6\} \to P(A \cap B) = 1/6$
- ▸ $P(A \mid B) = P(A \cap B) \;/\; P(B) = (1/6) \;/\; (3/6)$
- ▸ Simplify: $= 1/3$

**✓ Answer: $P(6 \mid \text{result} > 3) = 1/3 \approx 33\%$**

SOLUTION

# Q9: Normal Distribution — 68-95-99.7

**Problem: Scores ~ Normal($\mu$=70, $\sigma$=10). P(60 ≤ X ≤ 80)?**

**Step-by-step:**

- 60 and 80 are 10 units from the mean of 70.
- That's exactly 1 standard deviation: $10 = 1 \cdot \sigma$.
- So we need $P(\mu - \sigma \le X \le \mu + \sigma)$.
- By the 68-95-99.7 rule: about 68% of values are within $\pm 1\sigma$.

**✓ Answer: About 68%**

SOLUTION

# Q10: Binomial — Expected Value

**Problem: Spam filter catches 90% of spam. Out of 200 spam emails, expected catches?**

**Step-by-step:**

- Each email is a yes/no trial with $p = 0.9$ (catch).
- This is a Binomial setup with $n = 200$, $p = 0.9$.
- Expected value $= n \cdot p$
- $= 200 \cdot 0.9 = 180$

**✓ Answer: About 180 emails caught**

🎉

# Great work today!

*You now understand data types, descriptive stats, probability rules, conditional probability, and key distributions.*

Day 3 preview: Linear algebra — vectors, matrices, and how ML uses them.

**See you next class! 👋**